

DSP-based Dynamic Balancing Indicator

A newly developed digital system for detecting mass unbalance in rotating machinery is compact and easy-to-operate yet inexpensive



Developers with their product

ASHWIN GOPINATH

"Kee the wheel round for smoother operations and lower vibrations" was probably the first rule of wheel making, and rightly so.

Excessive vibration in rotating machinery can cause unacceptable levels of noise and also substantially reduce the life of shaft bearings. Hence the idea is to remove all causes of vibration to run the unit vibration-free. Unfortunately, in practice, this cannot be fully achieved and, in spite of the best designs, some inherent cause of vibration, or unbalance, will persist.

The smart thing to do is to reduce this unbalance to such a level that it will not adversely affect the bearing

life while reducing noise to an acceptable level.

The first step is to identify where the unbalances are and in what capacity. This is where a solution by Narayan Apte and Nikhil Mane comes in. Apte, a lecturer in electrical engineering at the Walchand College of Engineering, Sangli, and Mane, a lecturer in electronics engineering at KIT's College of Engineering, Kolhapur, have come up with a device aptly named 'digital electronic indicator for dynamic balancing of machines.' (I personally would have gone with 'DEI'.)

Mass unbalance

There are two centre lines of a rotating body: Physical centre line and rotating centre line. The mass of the rotating body is distributed around physical

centre line. If the mass distribution is even, both centre lines are coincident. But if the mass distribution is uneven, both centre lines are non co-incident. This gives rise to vibration in the shaft and subsequent problems arising out of the vibrations. Each rotating body needs to be checked for mass unbalance and should be corrected by removing or adding mass as required.

Conventional detection approach

The existing electronic hardware used for detecting mass unbalance is based on analogue circuitry. The indicator is based on an 8-bit microprocessor. Because of the analogue electronics involved, such devices have the drawbacks of difficult calibration, sensitivity to ambient parameter changes, large size, longer lead time (one to two weeks required for assembly and testing of one card) and higher cost.

The new approach

Apte and Mane developed a digital system based on the latest 32-bit processor from Texas Instruments (TI). The system is said to accurately detect the mass and angle of unbalance.

Apte explains the advantage of digital technology over analogue: "The existing electronic hardware was a combination of analogue and digital circuitry and hence it was quite bulky. Our system occupies negligible space and consumes much lesser power than the existing system. We are very confident that this device will have a lifespan of at least ten years."

The custom-developed digital electronic hardware is built around